HW 6.

1. Proof "if":

Assuming path from $x \to y$: $x \to x_1 \to x_2 \to \cdots x_{n-1} \to y$ (n steps),

then we got

$$\rho_{xy} = P(T_y < \infty \mid X_0 = x) \geq P(T_y \leq n \mid X_0 = x) \geq P(X_n = y \mid X_0 = x)$$

$$\geq P(X_n = y, X_{n-1} = x_{n-1}, \cdots X_1 = x_1 \mid X_0 = x_0)$$

$$= p(x_{n-1}, y) p(x_{n-2}, x_{n-1}) \cdots p(x_1, x_1) > 0$$

Therefore, we ~~proof~~ proved that $\rho_{xy} > 0$.

Proof "only if": If the HMC with transition probability P is irreducible, then for each pair $x_i, x_j \in C(P)$, there exists a path $x_i \to x_j$ and a path ~~x_j to x_j~~ x_j to x_i.

If there doesn't exist a path ~~from~~ from $x \to y$, indicating the MC starts at x cannot reach y, which means $T_y = \infty$, then we got $P(T_y < \infty \mid X_0 = x) = 0$, then the MC is not irreducible. This is a contradiction of the wanted situation. Therefore we ~~proof~~ proved the "only if" part by contradiction.

2. We have to show Δ is convex and $f(x) = P^T x$

For convexity of Δ,

for any $x, y \in \Delta$,

$$\sum_{i=1}^{S} x_i = \sum_{i=1}^{S} y_i = 1,\ x_i, y_i \geq 0,\ i = 1, 2, \cdots, S \quad (1)$$

$t \in (0,1)$, $z = tx + (1-t)y$ satisfy

$$\sum_{i=1}^{S} z_i = \sum_{i=1}^{S} (t x_i + (1-t) y_i) = t \sum_{i=1}^{S} x_i + (1-t) \sum_{i=1}^{S} y_i, \text{ because } (1),$$

$$= t + (1-t) = 1$$

$$z_i = t x_i + (1-t) y_i \geq 0,\ i = 1, 2, \cdots S$$

Therefore $z \in \Delta \Rightarrow \Delta$ is convex.

$\forall x \in \Delta$, let $y = f(x)$,

$$\sum_{i=1}^{S} y_i = \sum_{i=1}^{S} \left(\sum_{j=1}^{S} P_{ji} x_j \right) = \sum_{j=1}^{S} \left(\sum_{i=1}^{S} P_{ji} \right) x_j = \sum_{j=1}^{S} x_j = 1$$

$$y_i = \sum_{j=1}^{S} P_{ji} x_j \geq 0,\ i = 1, 2, \cdots, S$$

Thus $y \in \Delta$, $\Rightarrow f : \Delta \to \Delta$.

By the theorem, there exists $x \in \Delta$ such that $f(x) = x$,

$P^T x = x$, ✓

x is the stationary distribution π.

Assuming $\pi_k = (P^T x)_k = \sum_{j=1}^{S} P_{jk} \pi_j = 0$ ② Because $\pi_j \geq 0$, $P_{jk} \geq 0 \Rightarrow P_{jk} \pi_j \geq 0$ for all $j = 1, 2, \cdots, S$. For ② to work, it would need all elements are 0. Given $P_{jk} \pi_j \geq 0$, then π_j has to be 0 for all $j = 1, 2, \cdots S$, which is an impossible condition because we need $\sum_{j=1}^{S} \pi_j = 1$. Therefore, we prove the "Furthermore ... " in contradiction.

$$P = \begin{pmatrix} 0 & P_{12}\ 1/3 & P_{13}\ 2/3 \\ P_{21}\ 1/3 & 0 & 2/3\ P_{23} \\ P_{31}\ 1 & 0 & 0 \end{pmatrix}$$

(a) proof the MC is irreducible. According to the given P, $P_{12}, P_{13}, P_{21}, P_{23}, P_{31}$ all > 0, it indicating there is a path from X_1 to X_2, X_1 to X_3, X_2 to X_1, X_2 to X_3, X_3 to X_1. Then, we can combine path X_3 to X_1 and X_1 to X_2 to get the path for X_3 to X_2: $X_3 \to X_1 \to X_2$. Based on theorem 1.3, we successfully proved the MC is irreducible.

(b) Aperiodic

$(P^2)_{11} = \frac{7}{9} > 0$, $(P^2)_{22} = \frac{1}{9} > 0$, $(P^2)_{33} = \frac{2}{3} > 0$,

$(P^3)_{11} = \frac{2}{9} > 0$, $(P^3)_{22} = \frac{2}{9} > 0$, $(P^3)_{33} = \frac{2}{9} > 0$

$I_i = \{n \geq 1, (P^n)_{ii} > 0\} = \{2, 3\}$, 1 is not included in the set due to the fact that $(P^1)_{11} = 0$, $(P^1)_{22} = 0$, $(P^1)_{33} = 0$. The greatest common divisor is 1. Therefore, the MC is aperiodic.

(c) $P^T \cdot \pi = \pi$, P is the given P transition matrix,

$\sum_{i=1}^{3} \pi_i = 1$

$$\begin{cases} \pi_1 = \frac{1}{3}\pi_2 + \pi_3 \\ \pi_2 = \frac{1}{3}\pi_1 \\ \pi_3 = \frac{2}{3}\pi_1 + \frac{2}{3}\pi_2 \\ \pi_1 + \pi_2 + \pi_3 = 1 \end{cases} \Rightarrow \pi = \begin{pmatrix} 9/20 \\ 3/20 \\ 8/20 \end{pmatrix}$$

(d)

$$\sum_{x \in \mathcal{H}} f(x) \cdot \pi(x) = 1^2 \cdot \left(\frac{9}{20}\right) + 2^2 \cdot \left(\frac{3}{20}\right) + 3^2 \cdot \left(\frac{8}{20}\right) = \frac{93}{20}$$

(e) $\lim_{n \to \infty} (P^T)^n = (\pi \; \pi \; \pi)$

$$= \begin{pmatrix} \frac{9}{20} & \frac{9}{20} & \frac{9}{20} \\ \frac{3}{20} & \frac{3}{20} & \frac{3}{20} \\ \frac{8}{20} & \frac{8}{20} & \frac{8}{20} \end{pmatrix}$$

regardless of H's ~~initial~~ initial state, the long-term behavior of the MC:

$X_n \overset{!}{\sim} \pi$ ← Stationary Distribution

~~Approximately~~

$$\Rightarrow \lim_{n \to \infty} P^n = \left(\lim_{n \to \infty} (P^T)^n\right)^T$$

$$= \begin{pmatrix} \frac{9}{20} & \frac{3}{20} & \frac{8}{20} \\ \frac{9}{20} & \frac{3}{20} & \frac{8}{20} \\ \frac{9}{20} & \frac{3}{20} & \frac{8}{20} \end{pmatrix}$$